GEOLOGICAL NOTES

A PALEOMAGNETIC ESTIMATE OF THE AGE AND THERMAL HISTORY OF THE KENTLAND, INDIANA CRYPTOEXPLOSION STRUCTURE[1]

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ABSTRACT

The Kentland, Indiana, cryptoexplosion structure consists of a centrally uplifted zone of steeply-dipping fault blocks surrounded by a ring depression and a low-amplitude structural high with a radius of 12 km. The age of the deformation is constrained stratigraphically to the interval between Mississippian and Quaternary. Paleomagnetic samples were collected from seven carbonate formations and one siltstone unit. Of these only the Middle Ordovician Quimby's Mill Limestone contains a stable magnetization with consistent directions. Stepwise thermal and alternating-field demagnetization shows the presence of a northwesterly, moderately steep characteristic remanence with a median destructive field of 10 to 30 mT and maximum unblocking temperatures of 300°C. The pole position calculated from the mean direction falls near Late Cretaceous paleomagnetic poles. Dispersion increases when a bedding correction is applied, indicating post-deformational magnetization. Because the magnetization fails the fold test it is not a shock remanence. It seems likely, however, that its origin is related to the deformational event, which can thus be tentatively dated as Late Cretaceous. Acquisition of isothermal remanence indicates that magnetite is the only significant remanence-carrier in these rocks. Magnetic blocking theory suggests that the magnetization was acquired at temperatures of 150°C or less, over a timescale of approximately 1000 years. A thousand-year cooling time scale requires either continuous heating or a blanketing overburden of about 300 m, thus providing an estimate of the depth of subsequent erosion of the Kentland structure.

INTRODUCTION

Near Kentland, in northwestern Indiana (fig. 1), strongly tilted Ordovician and Silurian rocks occur in a complex domal structure (Schrock 1937; Gutschick 1961, 1976, 1982, 1983; Laney and Van Schmus 1978). The centrally uplifted zone consists of steeply-dipping fault blocks surrounding a core of highly shattered Lower Ordovician dolomites which have been displaced vertically by more than 600 m (Gutschick 1976, 1982). This uplifted portion is surrounded by a ring depression with a radius of 4 km in which Pennsylvanian channel deposits are preserved. This in turn is encircled by a low-amplitude structural high. The outermost limit of disturbance has a radius of 6 km (Laney and Van Schmus 1978). A large portion of the northern flank of the dome is exposed by the Newton County Stone quarry. The quarry, approximately 800 by 200 m in area with a maximum depth of 100 m, exposes a continuous, albeit faulted, stratigraphic sequence dominated by carbonates from the Lower Ordovician Shakopee Dolomite to the Middle Silurian Salamonie Dolomite.

Speculation and debate concerning the origin of the structure have gone on for a century. Gorby (1886) referred to the dome as a carbonate reef. Increased exposure due to quarrying operations in the 1920's made it clear that a major deformational event had taken place. A cryptovolcanic origin (Schrock 1937) emerged as the leading hypothesis but was challenged when Dietz (1947) published a study on the occurrence and orientation of shattercones, which favored a meteorite-impact origin. Dietz's proposal was supported by a report of the presence of coesite (Cohen et al. 1961) in the St. Peter Sandstone. A peak shock amplitude of 5 to 6 GPa (50 to 60 kbars) has been estimated by Laney and Van Schmus (1978). Polymictic and monomictic breccias as well as fault breccias occur within the quarry (Gutschick 1983), similar to those found in the proposed impact structures at Wells

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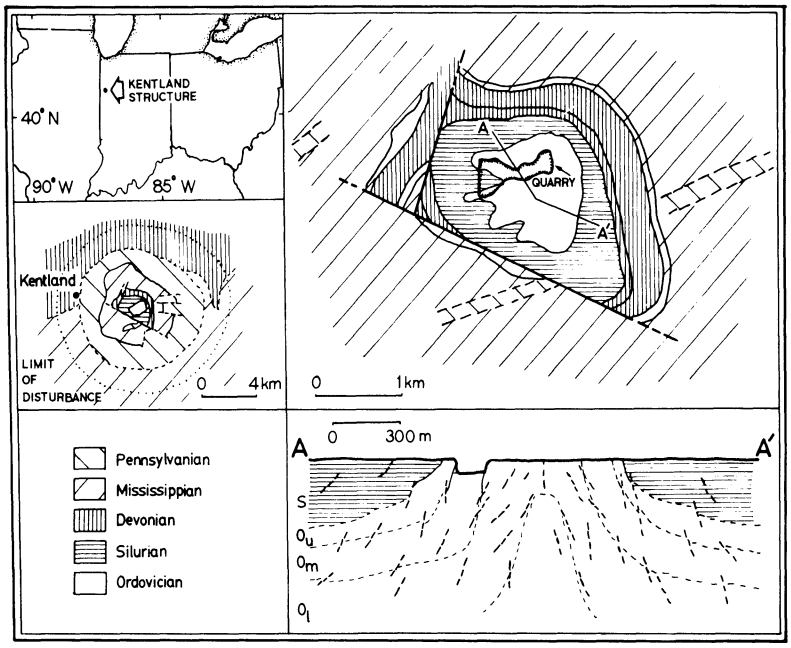


FIG. 1.—Large-scale geologic map at left (after Tudor 1971) shows concentric nature and extent of the Kentland structure. At right is a detailed geologic map (after Gutschick 1983), with the outline of the quarry, and a schematic cross section (after Laney and Van Schmus 1978) of the central part of the Kentland structure.

Creek, Tennessee (Wilson and Stearns 1968), Sierra Madera, Texas (Wilshire et al. 1971), and Decaturville, Missouri (Offield and Pohn 1979). The evidence for a high-velocity impact at Kentland has been summarized by Laney and Van Schmus (1978).

However, several objections have been raised to an impact origin for the structure. (1) Three-dimensional gravity modeling carried out by Tudor (1971) appears to require that the disturbance increases with depth, involving significant displacement of basement rocks. (2) Votaw (1980) reported a conodont color alteration index, or CAI (Epstein et al. 1977) of 1.5, which indicates anomalously low temperatures for an impact site (Gutschick 1983). (3) Similar breccias occur in intrusive structures such as the Monteregian carbonatite diatremes (Symons pers. comm.). (4) Endogenic processes have been proposed for formation of both shattercones (Sage 1978; Fleet 1979) and coesite (Chopin 1984). Thus, although high-velocity impact is currently the most widely accepted explanation for the origin of the Kentland structure, some uncertainty remains.

There is also considerable uncertainty regarding the age of the deformation. The overlying Quaternary cover is undeformed. The youngest sediments occurring in the quarry exposure are fragments of the upper Devonian New Albany Shale which occur in fault breccias (Gutschick 1983). Bedrock drill hole data (Tudor 1971) show that Mississippian rocks are folded and faulted; it is not clear whether the Pennsylvanian channel deposits were involved. The deformational event is thus no earlier than Mississippian, and older than the overlying Quaternary.

Paleomagnetic studies of several other known or suspected impact structures have been reported, including Barringer (Meteor) Crater, Arizona (Hargraves and Perkins 1969; Cisowski and Fuller 1978), Manicouagan, Quebec (Robertson 1967; Larochelle and Currie 1967), Ries, Germany (Pohl 1965), Rochechouart, France (Pohl and Soffel 1971), the Slate Islands, Lake Superior (Halls 1975, 1979), and Lonar Crater, India (Poornachandra Rao and Bhalla 1984). In the Manicouagan, Ries, and Rochechouart studies, the rocks sampled were breccias and glasses that had attained high temperatures and acquired a thermoremanent magnetization (TRM) during cooling. In the Slate Islands and Lonar studies, a characteristic pre-

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deformational magnetization was still present in the deformed rocks, but a secondary magnetization was present in the lower-coercivity fraction. For the Slate Islands rocks the intensity of the secondary magnetization decreased with radial distance from the center of the structure. A tilt correction improved the grouping of both the primary and secondary components of magnetization and also improved the statistical convergence of shattercone orientations. These observations argue for acquisition of a magnetic overprint during passage of an intense shock wave, termed shock remanent magnetization, or SRM (Halls 1975, 1979).

In addition, theoretical and experimental evaluations of the effects of shock on remanent magnetization have been reported by Hargraves and Perkins (1969), Nagata (1971), Pohl et al. (1975), and Cisowski and Fuller (1978). These studies confirm that shock waves with an amplitude of a few GPa are sufficient to impart a magnetization to rocks with a significant content of low- to moderate-coercivity magnetite. A majority of these studies dealt with igneous rocks, and only one (Cisowski and Fuller 1978) reported results on sedimentary carbonates. Kaibab Limestone samples from Barringer Crater contained a pre-deformational overprinted magnetization, which Cisowski and Fuller (1978) concluded was a SRM.

Cisowski and Fuller (1978) found that the range of unblocking temperatures (T_{ub}'s) of an experimentally-induced SRM extended beyond 500°C. The shock magnetization process thus appears to be controlled by the coercivity of the magnetite grains present, affecting only the low-coercivity fraction, but partially activating the entire T_{ub} spectrum. However, because T_{ub}'s depend upon coercivity as well as upon grain volume and time (Neel 1955), most of the SRM also resides in the low T_{ub} fraction and is therefore vulnerable to thermal perturbations. Thermal resetting of magnetic remanence at low to moderate temperatures (well below the Curie temperature) depends strongly on the time available for the magnetization to equilibrate with an ambient field. In natural events (impact or otherwise) where large volumes of rock are heated by shock, cooling proceeds slowly, and time may be available for significant thermal resetting of the SRM. In naturally shocked rocks which have attained moderate temperatures, one might expect any acquired SRM to be overprinted significantly by a post-deformational partial TRM in the low T_{ub} range, while any surviving SRM will reside in grains with higher T_{ub}'s.

This study was undertaken for two reasons. First, we believed we could obtain a paleomagnetic estimate of the age of the deformation at Kentland. Because of relatively rapid apparent polar wander (APW) since the Late Paleozoic, resolution of the age corresponding to a magnetic direction is quite good, and a paleomagnetic approach is well suited to the problem. Second, we planned to use the T_{ub} range of pre- and post-deformational components of magnetization to infer something about the shock temperatures reached in the sampling area.

METHODS

In a pilot study, oriented hand samples were collected from throughout the stratigraphic section. From two to four samples were obtained at two or three sites from each of eight exposed formations, mostly along the west wall of the quarry. All of these were carbonates except the Scales siltstone member of the Maquoketa Shale. Oriented cores were drilled in the laboratory and cut into standard 2.5 by 2 cm cylindrical specimens. Magnetization was measured on a 2-axis ScT cryogenic magnetometer. Following measurement of the natural remanent magnetization (NRM), specimens were subjected to stepwise alternating field (AF) or thermal demagnetization. Demagnetization data were plotted on orthogonal vector diagrams (Zijderveld 1967) and on stereographic projections, and visually inspected for segments of collinear or coplanar data points. The orientations of the best-fit lines or planes were determined by the principal components method of Kirschvink (1980). Mean directions were computed by the method of Bailey and Halls (1984). The pilot study indicated that only one of the formations, the Quimby's Mill Limestone, contained a stable magnetization with directional consistency. Results from the other formations will not be considered further here. In subsequent sampling a further 25 oriented cores of the Quimby's Mill Limestone were drilled in the quarry at four sites.

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RESULTS

NRM directions for the Quimby's Mill samples show a very consistent grouping around a northwesterly, fairly steeply downward direction (fig. 2), which becomes north-northwesterly and shallow after correction for the steep northerly dip of the bedding planes. Stepwise AF demagnetization most often showed small directional changes during progressive removal of the NRM, suggesting the presence of two magnetic components with overlapping AF stability (fig. 3). Commonly, a soft westerly component was preferentially removed, and in a few cases the direction of magnetization failed to reach a stable endpoint. The peak field required to reduce the initial NRM intensity by 50% (median destructive field, MDF) was typically between 10 and 30 mT. Treatment in the maximum available AF of 100 mT never left more than 10% of the initial NRM remaining. Incremental thermal treatments also showed a bivectorial decay of remanence in most cases (fig. 4). Unblocking temperatures were quite low: only one sample (JIQ 54, fig. 4) held a measurable remanence (intensity > 0.01 mA/m) after heating to 300°C, and no measurable magnetization remained in any sample after treatment at 330°C.

We interpret these two components of magnetization as an ancient natural remanence and a small superimposed drilling-induced remanence, or DIRM (Burmester 1977). In a previous study of DIRM in carbonate rocks (Jackson and Van der Voo 1985) we concluded that the remanence acquired during drilling is generally oriented parallel to the drill axis. Most of the cores were drilled in a westerly direction, in the plane of bedding, into the west wall of the quarry. These are the samples which contain a westerly magnetic overprint. About 10% of the cores were drilled in a southerly direction, normal to the bedding planes, and these samples show a southerly overprint. Comparison of the two groups shows that the DIRM is gen-

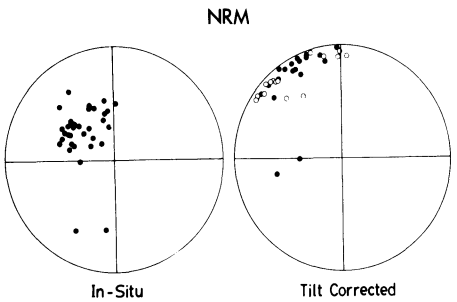


FIG. 2.—Equal area projections of the direction of natural remanent magnetization in samples of the Quimby's Mill Limestone from the Kentland quarry. Left, before correction for the tilt of the strata; right, after tilt correction.

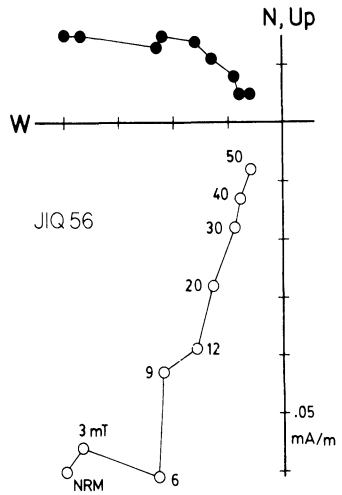


FIG. 3.—Demagnetization diagram (Zijderveld 1967) showing bivectorial decay of the natural remanence during alternating field demagnetization, for one specimen. Open symbols are projections of the successive positions of the endpoint of the magnetization vector onto the vertical (E,Up) plane; closed symbols are projections onto the horizontal plane. Points labeled in milliteslas.

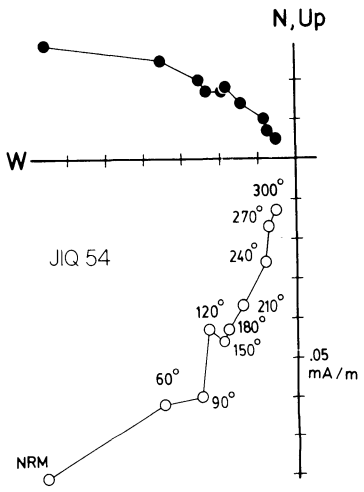


FIG. 4.—Results of thermal demagnetization, symbols as in figure 3. Temperatures in degrees Celsius. Note low unblocking temperatures.

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erally completely removed by heating to 150°C, or by AF cleaning to 15 or 20 mT. The characteristic remanence directions can then be resolved cleanly.

The characteristic directions fall in a tight group with a mean declination/inclination D/I = 314°/59°, with a Fisherian precision parameter k = 29.5 and a 95% confidence radius (α_{95}) = 4.8°. The site mean directions (fig. 5) have a mean D/I = 317°/62°, k = 142, α_{95} = 5.6° (N = 6). When a tilt correction is applied to the site mean directions, the dispersion increases significantly (fig. 5). The overall mean direction after correction is D/I = 336°/6°, k = 20.2, α_{95} = 15.3°. The fold test is conclusively negative, and thus the deformation predates the magnetization at the 99% confidence level (McElhinny 1964). The paleomagnetic pole position corresponding to the mean in situ remanence direction falls at 58° N, 198° E, close to the Late Cretaceous part of the North American APW path (fig. 6).

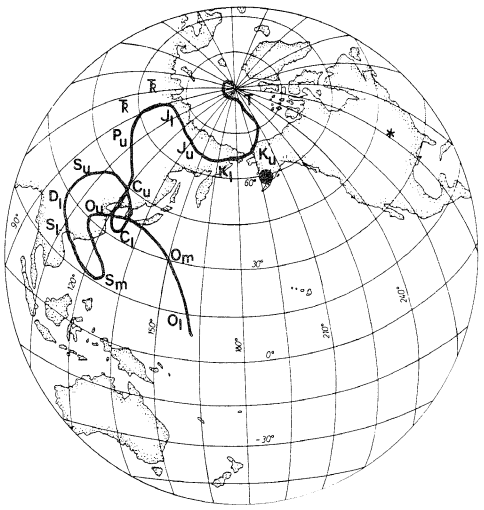


FIG. 6.—Paleomagnetic pole position for the Quimby's Mill Limestone at Kentland. The pole falls close to the late Cretaceous part of the apparent polar wander path of Van der Voo (1981), indicating a probable late Cretaceous age for the deformational event.

Isothermal remanent magnetization (IRM) acquisition curves (fig. 7*a*) approach saturation by 0.2 T and are nearly flat above 0.3 T. Such behavior is consistent with the remanence residing dominantly in magnetite (Dunlop 1973; Lowrie and Heller 1982). Thermal demagnetization of the saturation IRM (fig. 7*b*) shows that the Quimby's Mill samples are capable of holding a remanent magnetization at temperatures well above the maximum observed T_{ub} for the natural remanence, up to at least 400°C.

We conclude that the age of the characteristic magnetization in the Quimby's Mill Limestone samples is best estimated as Late Cretaceous, and the fact that all samples had normal polarity is consistent with this. Because the magnetization fails the fold test, it cannot be a SRM, and the deformational event must be older than the characteristic magnetization. Nevertheless it is reasonable to assume that they are related. Because the remanence resides exclusively in the lower T_{ub} fraction, the most likely explanation of its origin is that it was blocked-in during cooling immediately following the deformational event.

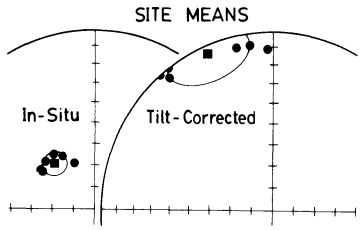


FIG. 5.—Site mean directions. Left, before and right, after correction for the bedding attitude. The magnetization fails the fold test at the 99% level of confidence.

THERMAL HISTORY

The low unblocking temperatures are consistent with the low conodont color alteration index of 1.5 observed by Votaw (1980), which indicates that high temperatures were not attained during the deformational event. Because the magnetization of an assembly of mineral grains approaches equilibrium with an external field in a time- and temperature-dependent manner, a magnetization which is unblocked in the laboratory during a 50-minute treatment may have been blocked-in at a considerably lower temperature over a longer time in nature. The time-temperature relations which govern magnetic blocking have been developed by Schmidt (1973), Pullaiah et al. (1975), Dodson and McClelland-Brown (1980), Walton (1980), and Dunlop (1981). Figure 8*a* shows the blocking relations of Walton (1980) for magnetite.

Conodont color alteration also proceeds

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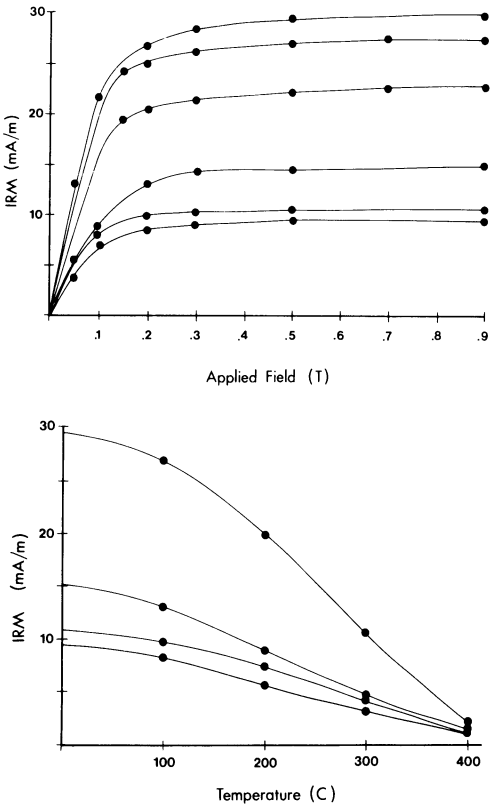


FIG. 7.—(*a*) Acquisition of isothermal remanent magnetization approaches saturation in fields of less than 0.3 tesla, indicating that magnetite is the dominant remanence carrier. (*b*) Thermal demagnetization of the saturated isothermal remanence shows that the samples are capable of holding a remanent magnetization up to 400°C or higher.

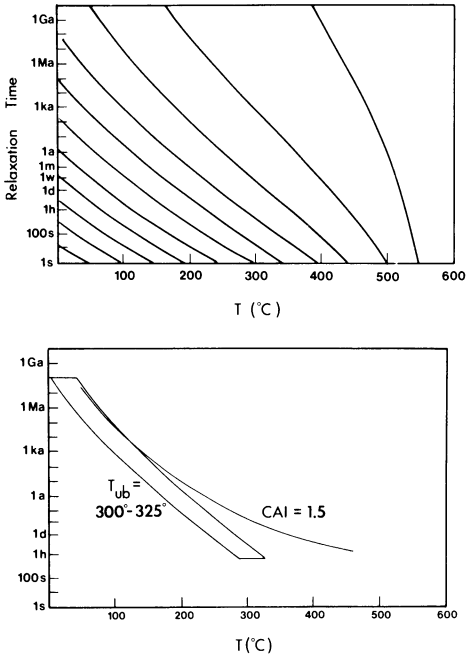


FIG. 8.—(*a*) Blocking relations of Walton (1980) for magnetite. (*b*) Time-temperature conditions for developing a conodont color alteration index of 1.5 (Epstein et al. 1977), compared with the conditions for acquisition of a remanent magnetization with a laboratory unblocking temperature of 300° to 325°C. The two sets overlap for temperatures less than 150°C and times longer than about 1000 years.

more rapidly at higher temperatures. The time and temperature dependence of this diagenetic process has been estimated by Epstein et al. (1977). Figure 8*b* shows the set of time and temperature conditions which give rise to a CAI of 1.5, compared with the conditions for development of a T_{ub} of 300–325°C. These separate sets of conditions overlap for temperatures less than about 150°C, maintained over an interval of more than about 1000 years. We infer that the deformational event was accompanied by relatively low temperatures; we will discuss the implications of a 1000-year time scale below.

It should be noted that the blocking relations we have used (Walton 1980) differ substantially from those proposed by other workers. In particular, this model involves a stronger time-dependence than the models of Pullaiah et al. (1975) and Dunlop (1981); i.e. the blocking contours in these other models are steeper than those shown in figure 8. Our inferences regarding the time and temperature involved in the deformational event are dependent on our choice of this magnetic blocking model. However, the experimental results of Pullaiah et al. suggest that their model underestimates the time dependence of the blocking process. Dunlop and Buchan (1977) concluded that the discrepancy between experimental results and the predictions of the more weakly time-dependent blocking model of Pullaiah et al. limited its quantitative applicability. Recently, Kent (1985) described a postdepositional component of remanence in glacial till with T_{ub}'s in excess of 300°C, in accord with the prediction of Walton's model. This model appears to agree more closely with available observational constraints, and we conclude that although there is considerable uncertainty in our estimate of the time and temperature involved in deformation of the Kentland rocks, it is the best estimate currently available.

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The question then arises: is such a low-temperature event compatible with a hypervelocity impact origin for the structure? Standard energy scaling relations of the form $KE \propto D^b$, where KE is the kinetic energy of the projectile immediately before impact, D is the crater diameter, and b is 3.4 to 3.6 (Dence et al. 1977), indicate that approximately 10^{18}–10^{19} J of energy were released in the Kentland area. Energy is partitioned during impact into kinetic and internal energies for the projectile and target. Model calculations for a gabbroic anorthosite target (O'Keefe and Ahrens 1977) have shown that 30% to 60% of the total energy of a 10–20 km/sec gabbroic or iron projectile goes into shock heating of the target. In a study of the effect of target material strength, Moore et al. (1963) found that, in weak rocks, there is less energy partitioned into target deformation and more into kinetic energy of ejecta, with the result that the fraction of energy partitioned into shock heating is independent of target strength (O'Keefe and Ahrens 1977). A hypervelocity impact would have emplaced about 10^{18} J of thermal energy within the Kentland rocks. If this were distributed uniformly throughout the roughly 10^{10} m^3 volume of the central uplift, it would raise the temperature by approximately 100°C. The actual distribution of energy depends on attenuation of the shock wave. If we use the estimate of Laney and Van Schmus (1978) for the shock amplitude experienced by the quarry rocks, we can make an approximate calculation of the corresponding local temperature increase.

Energy is deposited within a volume element during compression as a shock wave arrives. Pressure (P) and specific volume (V) evolve along a straight path in P,V space known as the Rayleigh line (Duvall 1962). The increase in internal energy at peak amplitude is given by the Rankine-Hugoniot energy conservation equation:

$$E = P_s(V_0 - V_s)/2 \tag{1}$$

where the subscripts 0 and s refer to the initial and shock-compressed states respectively (Duvall 1962). For peak shock pressures of 5 and 6 GPa, the deposited energy amounts to 2.54×10^5 and 3.30×10^5 J/kg, respectively, based on the limestone Hugoniot data of Ahrens and Gregson (1964). As the shock wave passes, the volume element re-expands, returning some of its internal energy to the shock wave. Pressure and volume return to their initial states along a release adiabat, which for solid materials is on the Hugoniot. The amount of residual energy remaining within the volume element is proportional to the area between the Rayleigh line and the Hugoniot (Duvall 1962). It is this residual energy that alters the conodonts and resets the magnetization.

We can define a Hugoniot by:

$$P = a^2\eta/V_0(1 - b\eta)^2 \tag{2}$$

(Gault and Heitowit 1963; Croft 1982), where the compression η is

$$\eta = 1 - V/V_0 \tag{3}$$

The constants a and b are derived from shock experiments on rocks and are obtained from a linear fit of shock velocity U and particle velocity u:

$$U = a + bu \tag{4}$$

From the data of Ahrens and Gregson (1964) for limestones, we have obtained values for a and b of 1400 m/sec and 2, respectively, for shocks up to 6.5 GPa. The net increase in internal energy following passage of the shock wave is then:

$$E_\eta = P_s(V_0 - V_s)/2 - a^2 \int_0^{\eta_s} \eta \, d\eta/(1 - b\eta)^2 \tag{5}$$

For peak pressures of 5 and 6 GPa, the net increase in internal energy amounts to 0.99×10^5 and 1.20×10^5 J/kg, respectively, corresponding to temperature increases of 99° and 120°C, assuming a specific heat of 1 kJ kg^{-1} K^{-1}. These temperatures are in quite good agreement with the estimates from magnetic blocking theory and conodont color alteration.

The timescale of 1000 years inferred from the magnetic blocking relations is quite long in view of the fact that the present depth of the sampling areas ranges from 25 to 50 m below ground surface. To maintain elevated temperatures over a span of 1000 years or

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more requires either continuous heating or an effective blanketing overburden. An approximate minimum overburden thickness can be obtained from a simple model in which the sampling area is instantaneously heated from temperature T_0 to T_1, and cooling takes place by conduction to the surface. The subsequent temperature history at depth z can then be calculated by the error function cooling law (Carslaw and Jaeger 1959):

$$(T - T_0)/(T_1 - T_0) = erf(z(4Kt)^{-1/2}) \quad (6)$$

where K is the thermal diffusivity.

Figure 9 shows the decay of anomalous temperature with time for various depths. The effect of surface cooling propagates downward. At 50 m the anomalous temperature is largely dissipated within 100 years, whereas at depths of a few hundred meters, it is over 100 years before cooling even begins. The persistence of high anomalous temperatures after initial cooling is an artifact of the half-space model. For our purposes, the significant information in figure 9 is the time when appreciable cooling begins at any depth. A timescale of 1000 years suggests the minimum postdeformational overburden thickness is approximately 300 m. An upper limit on the extent of post-deformational erosion is set by the maximum total thickness of post-Mississippian sediments found in the Upper Midwest region, approximately 2000 m (Willman et al. 1975).

Clearly this solution is valid only when lateral heat transport into and out of the local area is negligible in comparison to the vertical heat loss. The validity of the approximation deteriorates with increasing depth scales; however it is probably still reasonably accurate for a depth of 300 m, which is small in comparison with the lateral scale of the structure. It is interesting to note that Laney and Van Schmus (1978) also estimated an erosional depth of approximately 300 m by comparing styles of deformation with other proposed impact sites having undergone varying amounts of erosion.

DISCUSSION

Although the remanence in the Quimby's Mill Limestone samples from this study is not an SRM, we have shown that it can still be explained as a consequence of shock heating. Although we cannot rule out a cryptovolcanic energy source, we feel the agreement between the temperature calculated for a 5 GPa shock and that inferred from observed T_{ub}'s and CAI's provides circumstantial evidence in favor of a shock-related origin for the magnetization.

In previous studies of the effects of shock on remanent magnetization, thermal effects have been found to be important only for relatively strongly shocked (>20 GPa) rocks (Cisowski and Fuller 1978). Yet our results indicate that post-shock thermal effects were the dominant factor in the remagnetization of the Quimby's Mill Limestone in the Kentland quarry, where the shock level was only 5 to 6 GPa. We interpret the apparently anomalous nature of these results as primarily a conse-

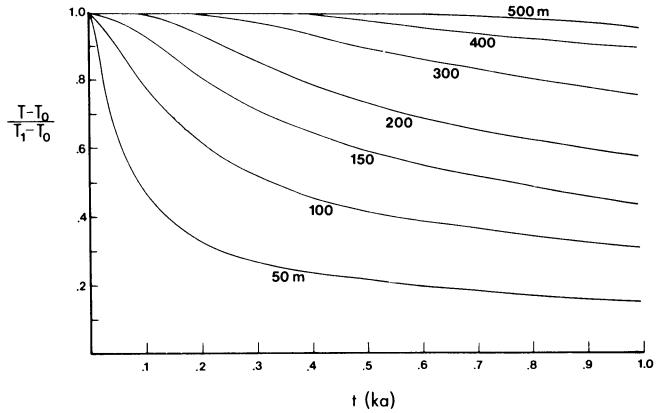


FIG. 9.—Error-function cooling model, for various burial depths (meters). At a depth of 50 m, the anomalous temperature is mostly eliminated within 100 years. A time scale of 1000 years indicates a probable depth of burial of at least 300 m.

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quence of lithology. In granitic and gabbroic rocks, approximately 20 GPa are required in order to attain a post-shock temperature of 150°C (Stoffler 1971). As previously mentioned, almost all documented cases of SRM involve rocks of this nature. Preservation of SRM appears to depend on the low post-shock temperatures maintained in such rocks. However, to account for the apparent survival of an SRM in the Kaibab Limestone (Cisowski and Fuller 1978) we must call on other factors. Because of the young age of Barringer crater (Sutton 1985), post-impact erosion has not been significant and the cooling time for the Kaibab samples, collected from the crater rim, was almost certainly very short. Insufficient time was available for thermal agitation to bring the magnetization toward equilibrium with the ambient field.

The hypothesis of a major impact or an increase in the rate of impact events coincident with the mass extinctions at the end of the Cretaceous (Alvarez et al. 1980; Emiliani et al. 1981) has received much recent attention. A Late Cretaceous age for the Kentland structure is therefore interesting, but probably not significant, since the uncertainty in our age estimate is at least 10 m.y. Shoemaker (1983) has estimated that the mean recurrence rate of impact events large enough to produce a 10 km crater is approximately 4/m.y. Therefore a more tightly constrained age estimate for the Kentland structure will have to be obtained before the significance of its proximity to the Cretaceous-Tertiary boundary can be assessed. Nevertheless, our results indicate that the Kentland structure should be added to the list of those considered potentially related to a terminal Cretaceous event.

SUMMARY AND CONCLUSIONS

The magnetization of samples of the Quimby's Mill Limestone from the Kentland structure appears to have been acquired during the late Cretaceous. It resides in magnetite, entirely in the low T_{ub} fraction, suggesting a thermal origin. The magnetization fails the fold test and is post-deformational at the 99% confidence level. We therefore believe that the magnetization was blocked in during cooling following structural displacement. The maximum T_{ub}'s are approximately 300° to 325°C. This, in conjunction with the low CAI (1.5) observed by Votaw (1980), indicates that high temperatures were not attained in the part of the structure now exposed in the quarry. Using the time-temperature relations of Walton (1980) for magnetic blocking, and Epstein et al. (1977) for conodont color alteration, we estimate that temperatures probably did not greatly exceed 150°C and that elevated temperatures were maintained over an interval of at least 1000 years. An error-function cooling model indicates that a minimum overburden thickness of about 300 m is required to maintain anomalous temperatures over such a time interval. Previous estimates of the shock level attained by the Kentland rocks average approximately 5 GPa. Our calculations using the limestone Hugoniot data of Ahrens and Gregson (1964) indicate that the residual temperature increase following passage of a 5 GPa shock wave is roughly 100°C. The thermal history we have inferred from observed unblocking temperatures appears to be entirely compatible with a shock origin for the structure.

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